Keeping documents readable for decades

FynePDF PDF/A bench

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Abstract

Archives keep documents for decades. This paper is a test file: it has sections that become bookmarks, mathematics, a table and, in its pdfLaTeX build, a photo and links.

1 Introduction

PDF/A fixes a document's appearance: every font travels with the file, colours are defined, and nothing depends on outside resources. For a document with n pages and f fonts, the work of embedding is $O(n + f)$, but the gain lasts:

$$\int_0^{\infty} e^{-\lambda t}\,\mathrm{d}t = \frac{1}{\lambda}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}. \tag{1}$$

2 Method

2.1 Files

We compare converters on the same files and measure validity, appearance and size.

Set	Files	Published
Feature	56	yes
Rules	1771	yes
Real world	1000	numbers only

Table 1: The file sets.

See `https://www.pdfa.org/` for the standard.

3 Conclusion

A converter must never claim conformance a file does not have. Greek letters α, β, γ and symbols $\leq, \geq, \infty$ must survive conversion.

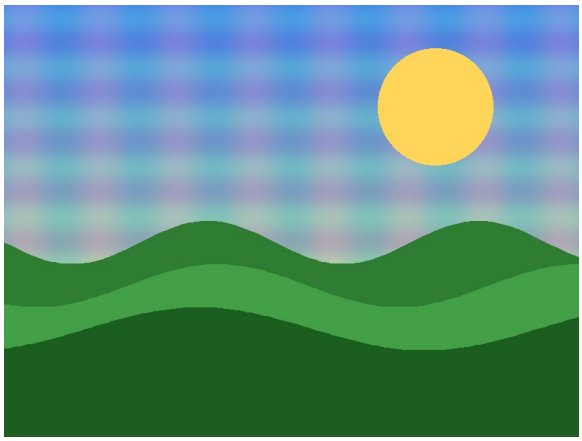


Figure 1: A photo, embedded as JPEG.